

Unit 2 – Exploring Two Variable Data

Notes 3 – Linear Regression

Recall

- If we think that a variable x may explain or even cause changes in another variable y , we call x an _____ and y a _____.
- A _____ displays the relationship between two quantitative variables measured on the same individuals.
- In examining a scatterplot, look for an overall pattern showing the _____, _____, and _____ of the relationship. And then for _____ or other deviations from the pattern.

Example

_____ Which of the following statements about correlation r is true?

- (A) A correlation of 0.2 means that 20% of the points are highly correlation.
(B) Perfect correlation, that is, when the point lie exactly on a straight line, results in $r = 0$.
(C) Correlation is not affected by which variable is called x and which is called y .
(D) Correlation is not affected by extreme values.
(E) A correlation of 0.75 indicates a relationship that is 3 times as linear as one for which the correlation is only 0.25

Least Squares Regression Line

Least Squares Regression or **linear regression** allows you to fit a line to a scatterplot in order to be able to better interpret the relationship between two variables, as well as make predictions about our response variable.

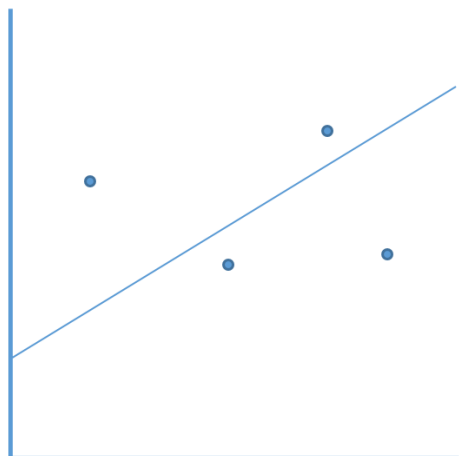
The fitted line is called the line of best fit, linear regression line, or least squares regression line, (LSRL) and has an equation in a form that should look very familiar:

	_____ : predicted y-value
	_____ : explanatory variable
	_____ : slope
	_____ : y-intercept

Another way the LSRL is written is _____

Slope is always the coefficient of x , and it may be called “a” or “b”

The way the line is fitted to the data is through a process called the **method of least squares**. The main idea behind this method is that the square of the vertical distance between each data point and the line is minimized.



- If you add all of the vertical distances from the point to the line, you get _____.
- We square each distance to make it positive, and then add it all up.
- *Not AP related: This is called the SSE (sum of the squared errors)*

Slope: The slope of the regression line is important in the sense that it gives us the change of y with respect to x . In other words, it gives us the amount of change in y when x increases by 1.

Intercept: The intercept is statistically meaningful only when x can actually take values close to zero. When it does make sense to have a x -value of zero, the y -intercept is the y -value we would expect.

When we have a data set (x,y) , we can calculate the LSRL by hand or with technology. In our Vitruvian Man activity, we were already introduced to both. Let's go through the process again, but this time, we will use a different example.

Example

Many schools require teachers to have evaluations done by students. A study investigated the extent to which student evaluations are related to grades. Teacher evaluations and grades are both given on a scale of 100. The results for Mrs. H for 10 of her students are given below together with the average for each student ($\bar{x}$).

x	40	60	70	73	75	68	65	85	98	90
y	10	50	60	65	75	73	78	80	90	95

(a) Create the LSRL by hand. Show all work.

Step 1: Enter the data into your calculator. x values go into L1 and y values into L2.

Step 2: Find the slope of your LSRL.

$$\text{Slope} = r \left(\frac{s_y}{s_x} \right)$$

$$r = \underline{\hspace{2cm}}$$

$$s_y = \underline{\hspace{2cm}}$$

$$s_x = \underline{\hspace{2cm}}$$

Slope =

*Finding r : We could go through the process of lists to find " r ", like we did in the activity, but we can just use our calculator to get us " r ". You will NOT have to find " r " by hand on the AP exam. You will have to show work for calculating the slope by hand.

*Also, when you find " r " on your calculator, yes you can see the LSRL from your calculator. Still show your work on your paper but feel free to make sure your answers match!

Step 3: Find the y-intercept of your LSRL.

$$\text{y-intercept} = \bar{y} - \text{slope} * \bar{x}$$

slope = _____

$\bar{x}$ = _____

$\bar{y}$ = _____

y-int = _____

*Fun fact of the LSRL, it will ALWAYS pass through the point $(\bar{x}, \bar{y})$. We use this property to find the y-intercept.

Step 4: Create your equation: _____ OR _____

(b) Use your equation to predict what evaluation Mrs. H will get from a student who scored a 81.

(c) Interpret the slope in the context of the problem.

(d) Do you think student grades and the evaluations students give their teachers are related? Explain.

LSRL on Calculator

There are two ways to get the line and it depends on how you like to write the line.

STAT → CALC → 4: LinReg(ax + b)

OR

STAT → CALC → 8: LinReg(a+bx)

Regression Line Cautions

Be careful of making predictions beyond what the data shows.

_____ is the use of a regression line for prediction far outside the interval of values of the explanatory variable x used to obtain the line. Such predictions are often not accurate.

CORRELATION DOES NOT IMPLY CAUSATION. For example, there was a study done that showed a strong, positive linear relationship between ice cream sales and homicides in New York City. Does this mean that if we stop selling ice cream, we will have no more homicides?

Coefficient of Determination

- The strength of a prediction which uses the LSRL depends on how close the data points are to the regression line.
- The mathematical approach to describing this strength is via the **coefficient of determination** (r^2).
- The coefficient of determination gives us the proportion of variation in the values of y that is explained by least-squares regression of y on x.
- The coefficient of determination turns out to be the correlation coefficient squared.

In our last example, $r =$ _____ Therefore, to find the coefficient of determination:

Whenever you use the regression line for prediction, also include as a measure of how successful the regression is in explaining the response.

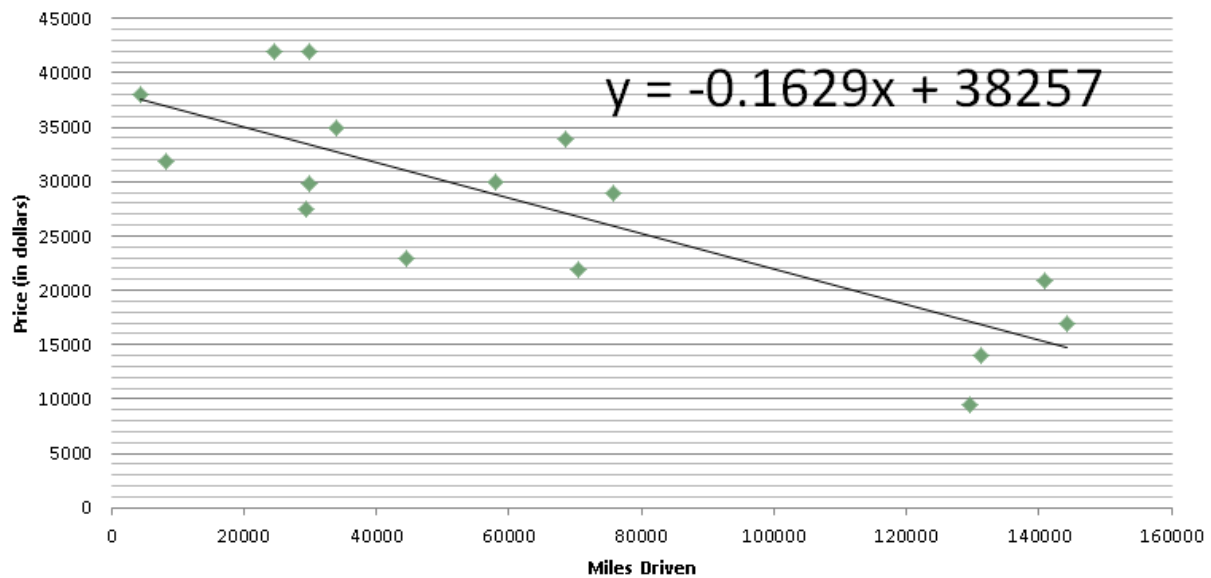
This means that _____% of the variation in teacher evaluations (the dependent variable) can be explained by the linear relationship it has with the student class average (the independent variable).

Residuals

- In most cases, no line will pass exactly through all the points. This means that even if we use the LSRL to make predictions about our dependent variable, there will still be some error from the actual y-value.
- Because we use the line to predict y from x, the prediction errors we make are errors in y, the _____ direction in the scatterplot.
- A good regression makes the vertical deviations of the points from the line _____ (remember LEAST Square Regression)
- A residual is the _____ between an observed value of the response variable and the value predicted by the regression line.
- Residual = observed – predicted OR _____
- If the residual is positive, the observed point lies _____ the least squares regression line.
- If the residual is negative, the observed point lies _____ the least squares regression line.

Fun fact: If you add up all the residuals from your data, you will get “0”. That is why the method of LSRL involves squaring the residuals then adding them up and minimizing that value.

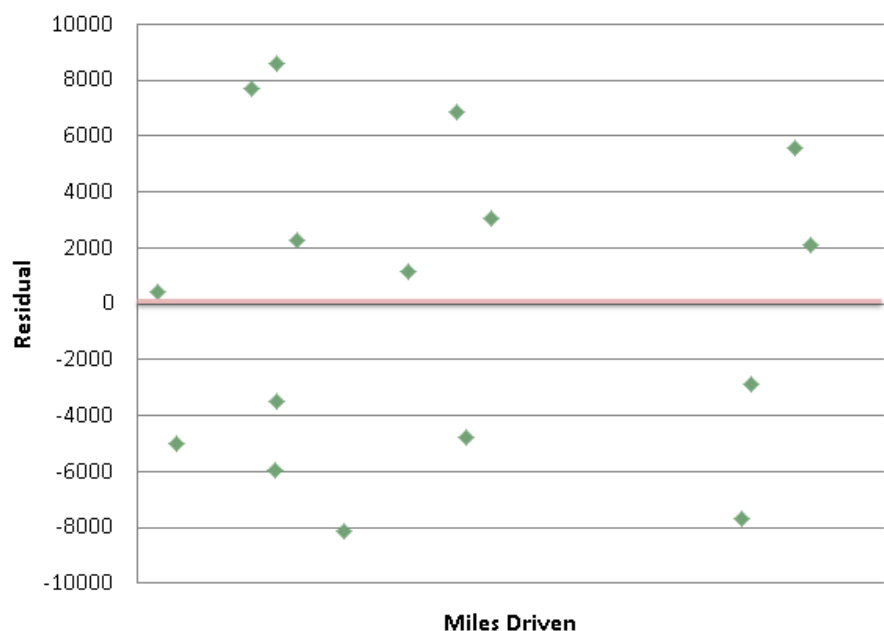
Example: Everyone knows that cars and trucks lose value the more they are driven. Can we predict the price of a used Ford F-150 SuperCrew 4x4 if we know how many miles it has on the odometer? A random sample of 16 used trucks was selected from autotrader.com. Here is a graph of the data:



Find and interpret the residual for the Ford F-150 that had 70,583 miles driven and a price of \$21,994.

Residual Plots

- A residual plot makes it easy to study the residuals by plotting them against the explanatory variable.
- Residual plots help us assess whether a linear model is appropriate.
- In the truck example, if we calculate all the residuals for each point, we can then plot the miles driven vs. residuals.



- Essential, a residual plot turns the regression line horizontal.
- Residual plots magnify the deviations of points from a line.
- This makes it easier to see an unusual pattern, so it helps us determine if a linear model is appropriate.

When an obvious pattern exists in a residual plot, the model we are using is not appropriate:

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Residual Plots on Calculator

The TI-83/84 will generate a complete set of residuals when you perform a `LinReg`. They are stored in a list called `RESID` which can be found in the `LIST` menu. `RESID` stores only the current set of residuals. That is, a new set of residuals is stored in `RESID` each time you perform a new regression.

In order to draw a residual plot on the TI-83/84, first enter your data and perform a `LinReg`. Next, create a `STAT PLOT` where `XList` is `L1` and `YList` is `RESID` (get this by pressing `2nd → STAT → 9:RESID`)